

Chapitre 11 : Tableaux

Solution exercice

Exercice 1

```
public class PremiersEntiers {
    public static void main(String[] args) {
        int [] carre = new int [10];

        for (int i = 0; i < carre.length; i++)
            carre[i] = i * i;

        System.out.print("Premiers entiers au carré : ");

        for (int i = 0; i < carre.length; i++)
            System.out.print( carre[i] + " ");

    }
}
```

Exercice 2

```
import javax.swing.JOptionPane;

public class JoursSemaine {
    public static void main(String[] args) {
        String jours[] = {"Dimanche", "Lundi", "Mardi", "Mercredi", "Jeudi", "Vendredi", "Samedi"};
        String joursRaccourcis[] = new String [jours.length];
        String texteAff = "";

        for (int i = 0; i < jours.length; i++){
            joursRaccourcis[i] = jours[i].substring(0, 3);
            if (i == 0)
                texteAff += joursRaccourcis[i];
            else
                texteAff += ", " + joursRaccourcis[i];
        }

        JOptionPane.showMessageDialog(null, texteAff);
    }
}
```

```
}  
}
```

Exercise 3

```
public class ListeEntiers {  
    private int entiers[];  
  
    public ListeEntiers(String liste) {  
        String listes [] = liste.split(" ");  
  
        entiers = new int[listes.length];  
  
        for (int i = 0; i < listes.length; i++){  
            entiers[i] = Integer.parseInt(listes[i]);  
        }  
    }  
  
    public String toString() {  
        String txt = "";  
  
        for (int i = 0; i < entiers.length; i++){  
            txt += "Tableau[" + i + "] = " + entiers[i] + "\n";  
        }  
  
        return txt;  
    }  
  
    public String afficheTab(){  
        String txt = "";  
  
        for (int i = 0; i < entiers.length; i++){  
            txt += entiers[i] + " ";  
        }  
  
        return txt;  
    }  
}
```

```

    public int somme(){
        int somme = 0;

        for (int i = 0; i < entiers.length; i++){
            somme += entiers[i];
        }

        return somme;
    }

    public int moyenne(){
        int somme = 0;

        for (int i = 0; i < entiers.length; i++){
            somme += entiers[i];
        }

        return somme/entiers.length;
    }
}

```

Exercise 4

```

public class ListeMots {

    private String mots [];

    public ListeMots(String liste) {
        mots = liste.split(" ");
    }

    public String longueursMots(){
        String txt = "";

        for (int i = 0; i < mots.length; i++){
            txt += mots[i] + "\t" + mots[i].length() + "\n";
        }
    }
}

```

```

        return txt;
    }

    public static void main(String[] args) {
        ListeMots maListe = new ListeMots("Bonjour le monde");

        System.out.println(maListe.longeursMots());
    }
}

```

Exercice 5 et 6

```

public class ListeEntiers {

    private int entiers[];

    public ListeEntiers(String liste) {
        String listes [] = liste.split(" ");

        entiers = new int[listes.length];

        for (int i = 0; i < listes.length; i++){
            entiers[i] = Integer.parseInt(listes[i]);
        }
    }

    public String toString() {
        String txt = "";

        for (int i = 0; i < entiers.length; i++){
            txt += "Tableau[" + i + "] = " + entiers[i] + "\n";
        }

        return txt;
    }

    public String afficheTab(){

```

```
        String txt = "";

        for (int i = 0; i < entiers.length; i++){
            txt += entiers[i] + " ";
        }

        return txt;
    }

    public int somme(){
        int somme = 0;

        for (int i = 0; i < entiers.length; i++){
            somme += entiers[i];
        }

        return somme;
    }

    public int moyenne(){
        int somme = 0;

        for (int i = 0; i < entiers.length; i++){
            somme += entiers[i];
        }

        return somme/entiers.length;
    }

    public int posMaximum(){
        int valeurMax;
        int positionDuMax;

        positionDuMax = 0;
        valeurMax = entiers[0];

        for (int i = 1; i < entiers.length; i++)
        {
            if (entiers[i] > valeurMax)
            {
                positionDuMax = i;
                valeurMax = entiers[i];
            }
        }
    }
}
```

```
    }

    return positionDuMax;
}

public int maximum(){
    return entiers[this.posMaximum()];
}

public int posMinimum(){
    int valeurMin;
    int positionDuMin;

    positionDuMin = 0;
    valeurMin = entiers[0];

    for (int i = 1; i < entiers.length; i++)
    {
        if (entiers[i] < valeurMin)
        {
            positionDuMin = i;
            valeurMin = entiers[i];
        }
    }

    return positionDuMin;
}

public int minimum(){
    return entiers[this.posMinimum()];
}

public int recherche(int valeurCherche){

    for (int i = 0; i < entiers.length; i++)
    {
        if (entiers[i] == valeurCherche)
        {
            return i;
        }
    }

    return -1;
}
```

```
    }  
}
```

Exercise 7

```
public class Conjugaison {  
    private String verbe;  
    private String pronoms[] = {"je", "tu", "il/elle", "nous", "vous", "ils/elles"};  
    private String terminaisons [] = {"e", "es", "e", "ons", "ez", "ent"};  
  
    public Conjugaison(String verbe) {  
        this.verbe = verbe;  
    }  
  
    public String getVerbe() {  
        return verbe;  
    }  
  
    public void setVerbe(String verbe) {  
        this.verbe = verbe;  
    }  
  
    public boolean estVerbeER(){  
        if(verbe.substring(verbe.length()-2).equals("er"))  
            return true;  
        else  
            return false;  
    }  
  
    public String afficheTable(){  
        String txt = "";  
        for(int i = 0; i < pronoms.length; i++)  
            txt += pronoms[i] + " " + verbe.substring(0, verbe.length() - 2) + terminaisons[i] + "\n";  
        return txt;  
    }  
}
```

Exercise 8

```
public class MotsMysteres {
    private String listeMots[] = {"rose", "rouge", "vert", "bleu", "jaune"};

    public boolean dansListe(String mot){
        for (int i = 0; i < listeMots.length; i++)
        {
            if (listeMots[i].equalsIgnoreCase(mot))
            {
                return true;
            }
        }

        return false;
    }
}
```

Exercise 9

```
public class JeuCapitales {
    private String pays [] = {"ALLEMAGNE", "AUTRICHE", "BELGIQUE", "BULGARIE", "CANADA", "CHILI",
    "CHINE", "COLOMBIE", "COREE-DU-SUD", "CUBA", "DANEMARK", "ESPAGNE", "ETATS-UNIS", "FINLANDE", "FRANCE",
    "GRECE", "HONGRIE", "IRLANDE", "ITALIE", "JAPON"};
    private String capitales [] = {"BERLIN", "VIENNE", "BRUXELLES", "SOFIA", "OTTAWA", "SANTIAGO",
    "PEKIN", "BOGOTA", "SEOUL", "HAVANE", "COPENHAGUE", "MADRID", "WASHINGTON", "HELSINKI", "PARIS",
    "ATHENES", "BUDAPEST", "DUBLIN", "ROME", "TOKYO"};

    private int paysMistere;

    public JeuCapitales() {
        this.genererPaysMistere();
    }

    public String getPays() {
        return pays[paysMistere];
    }
}
```

```
public String getCapital() {  
    return capitales[paysMistere];  
}  
  
public void genererPaysMistere(){  
    paysMistere = (int) Math.floor(Math.random() * pays.length);  
}  
  
public boolean aTrouver(String capitale){  
    if(capitale.equalsIgnoreCase(capitales[paysMistere])){  
        return true;  
    }  
    else{  
        return false;  
    }  
}  
}
```

Exercice 10

```
160
```

Exercice 11

```
tab[0] = 28
```

Exercice 12

```
2 63
```

Exercice 13

```
Bonjour  
Bonjour  
2 4
```

Exercice 14

```
15  
30  
  
Somme = 30
```